

Neural-Symbolic Post-Tonal Composition Assistance with Pitch-Class Set, Serial, and Rhythmic-Profile Constraints

融合音级集合、序列与节奏轮廓约束的神经符号后调性作曲辅助方法

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Abstract

Explicit control of post-tonal material remains underrepresented in symbolic music generation, while evaluation on copyrighted contemporary repertoire is difficult to release and reproduce. We present a score-level neural-symbolic composition assistant conditioned on pitch-class sets and interval vectors, optional twelve-tone rows and serial transformations, rhythmic profiles, gestures, instrumentation, voice count, and requested score span. A legal procedural generator supplies an explicit corpus split of 20,000 training, 2,000 validation, and 2,000 test fragments. A causal Transformer learns condition-preserving event windows, and inference-time symbolic analysis reranks sampled candidates before MusicXML and a JSON report are exported. The corrected evaluation separates serial row targets from non-serial pitch-class-set targets, hides stochastic density targets from decoding, evaluates condition ablations against the original held-out targets, and checks requested measure and voice counts in exported scores. Corrected full neural metrics are PENDING_REAL_EXPERIMENT until every configured checkpoint and evaluation gate completes. The present manuscript therefore specifies the auditable method and claim boundaries without substituting smoke or legacy results for the formal experiment.

Keywords: computer-assisted composition; post-tonal music; symbolic music generation; pitch-class set theory; twelve-tone serialism; constraint-guided decoding; MusicXML

1 Introduction

Post-tonal composition frequently begins from explicit symbolic materials: a pitch-class collection, an interval-class profile, a twelve-tone row and its transformations, a registral trajectory, or a characteristic rhythmic surface. Computer-assisted composition environments have long supported the manipulation of such structures, but recent neural music systems are more often evaluated on tonal continuation, popular-song structure, performance MIDI, or audio generation [2, 3]. Those settings do not directly answer a practical question faced by composers of contemporary score-

based music: can a generative model produce an editable sketch while preserving named post-tonal constraints and explaining where it satisfies or violates them?

This question requires a different interface and a different evaluation vocabulary. Pitch-class set theory supplies normal order, prime form, and interval vectors [6, 13]; serial practice requires explicit $P/R/I/RI$ transformations and aggregate tracking; rhythmic and gestural intentions are expressed over score time rather than audio frames. The output must also be inspectable as notation. A fluent token sequence is therefore insufficient if it cannot be converted into a score or if its relation to the requested material remains opaque.

Data legality presents a second challenge. A large fraction of relevant post-1945 repertoire remains copyrighted, and a scraped corpus would be difficult to release or reproduce. We instead use a deterministic procedural corpus as a controlled testbed. The corpus is not intended to imitate a composer or establish historical style. Its purpose is to expose known condition–target relations so that constraint adherence, failure modes, and decoding trade-offs can be measured without distributing protected scores.

We study a condition-prefix Transformer that generates quantized event sequences and a neural-symbolic decoder that reranks sampled candidates using non-differentiable post-tonal diagnostics. The approach is deliberately modest at the neural level: it does not introduce a new attention architecture. Its contribution lies in connecting causal symbolic generation to pitch-class-set, serial, rhythmic, gestural, register, and notation-level checks, and in returning both a MusicXML sketch and a machine-readable explanation.

The corrected study addresses three research questions:

- **RQ1:** With one trained generator and fixed test conditions, how does four-candidate symbolic reranking change explicit constraint metrics relative to its aligned first candidate?
- **RQ2:** How does hiding pc-set, serial, rhythm, or gesture fields from the condition prefix affect adherence to the unchanged held-out targets?
- **RQ3:** Can generated fragments be exported as structurally valid MusicXML with the requested measures and voices and with inspectable analysis reports?

The main contributions are:

1. a legal and reproducible 24,000-fragment synthetic corpus protocol for score-level post-tonal conditions;
2. a compact condition-prefix/event representation and locally trainable Transformer conditioned on requested spans of 4–16 measures and 2–8 voices;
3. an inference-time candidate-reranking objective grounded in pc-set, serial, rhythmic, gestural, and range diagnostics; and

4. a provenance-gated evaluation and export layer that connects aggregate metrics to MusicXML examples and per-fragment JSON reports.

The resulting claims are intentionally bounded. Corrected quantitative findings will concern a synthetic constraint testbed, not stylistic authenticity. Expert judgments of compositional usefulness are planned with the supplied blind-rating package but are not inferred from automatic metrics. Until the formal v3 run completes, numerical neural claims are marked `PENDING_REAL_EXPERIMENT`.

2 Related Work

2.1 Computer-assisted composition and post-tonal representation

Computer-assisted composition has a longer history than data-driven music generation. OpenMusic represents musical objects and transformations in a visual programming environment, allowing a composer to formalize and revise compositional procedures without reducing the task to audio synthesis [2]. The music21 toolkit provides a complementary programmatic layer for symbolic score analysis, transformation, and interchange [4]. These systems establish symbolic inspectability as a practical requirement for compositional software.

Constraint programming offers a second precedent for explicit musical control. Anders and Miranda survey systems that express theories of harmony, rhythm, form, and instrumentation as declarative constraints [1]. This tradition treats a musical rule as an inspectable compositional object rather than as an implicit property of a corpus. Our decoder follows the same preference for explicit diagnostics, while using sampled neural candidates instead of solving a complete constraint-satisfaction problem.

Pitch-class set theory supplies equivalence under transposition and inversion, normal and prime forms, and interval-class vectors for non-tonal pitch organization [6, 13]. Post-tonal theory also treats ordered twelve-tone rows, $P/R/I/RI$ transformations, and aggregates as distinct from unordered set classes. The present system keeps that distinction in both its condition representation and its analysis report.

2.2 Neural symbolic music generation

Neural music generation spans different objectives, representations, and forms of control [3]. The Transformer architecture provides a standard causal sequence model [14], and Music Transformer adapted self-attention to long symbolic performance sequences [10]. Later event representations made meter and beat structure more explicit for expressive pop-piano generation [11]. These studies show that event sequences can support long musical outputs, but their principal corpora

and evaluation tasks do not directly measure pitch-class-set adherence, serial transformations, or post-tonal gesture control.

Other symbolic models emphasize polyphonic coordination or latent structure. MuseGAN generates several instrumental tracks in a piano-roll representation and evaluates a rock-music accompaniment setting [5]. MusicVAE uses a hierarchical recurrent decoder to model longer musical sequences in a continuous latent space [12]. These objectives are relevant to multi-part organization and variation, but neither formulation exposes the pitch-class-set and row-form conditions required by the present task.

Our model therefore uses the Transformer as an established sequence generator rather than claiming an architectural advance. Its event vocabulary is intentionally smaller than performance-oriented encodings because the output target is a quantized, editable score fragment. Conditions for pitch, serial order, rhythm, gesture, instrumentation, and span are written into the same symbolic sequence as a prefix.

2.3 Constraint-aware generation

Several systems make neural generation steerable. DeepBach permits positional constraints on notes, rhythms, and cadences during pseudo-Gibbs sampling [8]. Anticipation-RNN incorporates future unary constraints into sequence generation through a backward constraint network and a forward token network [7]. Outside purely neural sampling, MorpheuS uses optimization to preserve recurrent patterns and follow a target tension profile [9]. These methods motivate explicit control, but they address different musical domains and constraint types.

FIGARO is a closer precedent for interpretable conditional generation. It reconstructs symbolic music from learned and expert descriptions, including time-varying high-level features [15]. Our study narrows the control vocabulary to post-tonal compositional materials and adds deterministic analysis after sampling. The resulting comparison asks whether reranking candidates by pitch-class, serial, rhythmic, gestural, and range diagnostics changes adherence while the trained checkpoint and test requests remain fixed.

The approach studied here separates probabilistic generation from deterministic score analysis. A causal Transformer samples several candidates, after which post-tonal functions rerank them by pitch-class coverage, interval-vector distance, row order, aggregate completion, rhythmic profile, gesture features, and instrumental range. This design accepts the computational cost of multiple candidates in exchange for diagnostics that can be shown to a composer and saved with the score.

2.4 Position of the present study

The study connects three established practices: formal symbolic manipulation from computer-assisted composition, event-based neural sequence generation, and constraint-aware selection. Its

narrower contribution is a workflow conditioned on 4–16-measure post-tonal sketch requests that returns both MusicXML and an explanation of constraint adherence. The legal synthetic corpus makes the condition–target relation reproducible. The corrected experimental protocol is designed to test the contribution of constraint-guided selection without changing the checkpoint or test material within each paired comparison; numerical conclusions remain pending until that protocol completes.

3 Methodology

3.1 Task formulation

The system generates a quantized symbolic score fragment x from a condition bundle c . The bundle may contain a pitch-class set S , a six-entry interval vector $v(S)$, a twelve-tone row r , a serial form $f \in \{P_n, R_n, I_n, RI_n\}$, a rhythmic profile, a gesture label, an instrument, a voice count, and a measure count. The requested output contains 4–16 measures and 2–8 parts and is returned as event tokens, MusicXML, and a score-level analysis report. No waveform or audio representation is used.

The conditional language model factorizes

$$p_\theta(x \mid c) = \prod_{t=1}^T p_\theta(x_t \mid c, x_{<t}), \quad (1)$$

where x_t is a score-event token. The research objective is to combine next-token modeling with explicit, inspectable post-tonal constraints.

3.2 Legal procedural corpus

Because much post-1945 art-music repertoire remains copyrighted, the bundled corpus is generated procedurally rather than scraped from published scores. Each sample specifies 4–16 measures, 2–8 voices, one instrument label, one of seven rhythmic profiles, and one of seven gesture labels. Non-serial samples receive a pitch-class set and interval vector. Serial samples instead receive a valid twelve-tone row and one of 48 transposed $P/R/I/RI$ forms. This separation prevents an impossible target in which one passage is simultaneously required to remain in a small set and complete the chromatic aggregate.

The rule generator creates a rhythmic skeleton for each voice, applies gesture-dependent onset, duration, rest, and register operations, and selects pitches inside the requested instrument range. Time is quantized to 0.25 quarterLength. The random density curve realized in each target is saved for held-out evaluation, but it is removed from model-visible condition metadata and is not used by candidate reranking.

3.3 Condition and event representation

Conditions are serialized as a prefix beginning with **BOS** and ending with **SEP**. The prefix can include pitch-class, interval-vector, row-pitch, row-form, rhythmic-profile, gesture, voice-count, measure-count, and instrument tokens. Explicit **NO_PCSET**, **NO_ROW**, **NO_PROFILE**, and **NO_GESTURE** symbols identify absent conditions.

The score body contains **BAR**, **TIME_SHIFT**, **VOICE**, **PITCH**, **OCTAVE**, **DUR**, and **REST** events followed by **EOS**. A finite-state grammar constrains decoding order, requested voice indices, instrument range, and requested bar count. The parser clamps decoded events to the requested span.

Many complete fragments exceed the 256-token model context. Training therefore uses windows that preserve the conditions instead of silently truncating the sequence. Every window repeats the complete condition prefix and retains preceding score context when capacity permits. Loss is applied only to the current score-body target segment. A ten-epoch coverage cycle partitions the target windows so that every saved score-body token contributes once per cycle. Validation and teacher-forced evaluation enumerate all windows. Reported token accuracy and cross-entropy are body-token diagnostics, not measures of compositional quality.

3.4 Conditional Transformer

The generator is a causal Transformer [14] with six layers, hidden dimension 384, six attention heads, GELU feed-forward blocks of width $4d$, learned positional embeddings, pre-normalization, and dropout 0.1. The training objective is teacher-forced negative log likelihood over the unmasked score-body labels:

$$\mathcal{L}_{\text{NLL}}(\theta) = - \sum_{t \in \mathcal{T}} \log p_{\theta}(z_t \mid z_{<t}), \quad (2)$$

where $\mathcal{T}$ denotes the current target window. No pretrained language model or external score corpus is used.

3.5 Constraint-guided candidate reranking

At inference, the proposed decoder samples K grammar-constrained candidates and analyzes each at score level. Candidate selection is

$$x^* = \arg \min_{x \in \mathcal{C}_K} \sum_j \lambda_j m_j(x, c), \quad (3)$$

where m_j includes missing pc-set coverage, off-set pitch classes, interval-vector distance, cyclic row-order error, incomplete strict serial forms, incomplete serial aggregates, rhythmic-profile distance, gesture inconsistency, instrument-range violations, insufficient content span, and missing requested voices. Pitch-class terms are inactive without a pc-set target, and serial-form and aggregate terms are inactive without a row target. The held-out stochastic density curve is excluded from this

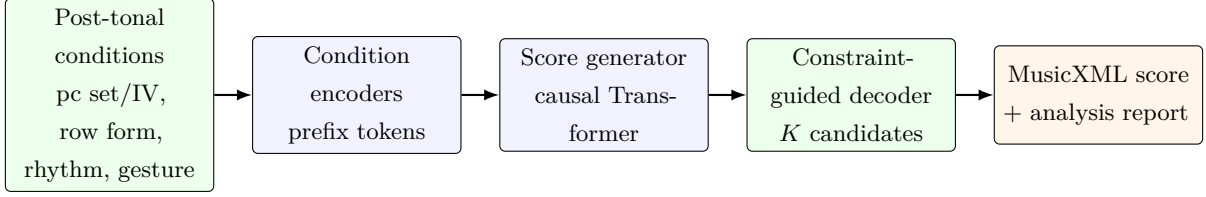


Figure 1: Corrected inference path. Post-tonal conditions are serialized by condition encoders, consumed by the causal score generator, and evaluated by a constraint-guided decoder before MusicXML and an analysis report are exported. A legal procedural generator supplies the condition–score pairs used for training.

objective. These non-differentiable terms are used only for candidate selection and are not presented as training losses.

The primary controlled contrast uses one shared trained generator: $K = 1$ is the vanilla decoder and $K = 4$ is the guided decoder. A deterministic random-number stream is assigned to each test condition, making the first $K = 4$ candidate identical to the corresponding $K = 1$ output. Condition-removal models use the same corpus samples and target events while modifying only model-visible prefix fields. Evaluation retains the original hidden targets.

3.6 Analysis and MusicXML export

Each selected fragment receives pc-set coverage, pc-set precision and Jaccard similarity, generated interval vector, interval-vector distance, cyclic row-order accuracy, strict complete-form accuracy, serial aggregate completion, density curve, rhythmic-profile distance, held-out density-curve error, gesture features, gesture consistency, register spread, range violations, content-span adherence, and requested-voice adherence. Strict serial-form accuracy counts complete non-overlapping twelve-note blocks and is independent of cyclic row-order accuracy.

MusicXML export creates the requested number of score parts and measures, pads empty trailing measures, and removes events outside the requested span [16]. Success is recorded separately for structural parsing, requested measure count, and requested part count.

4 Experimental Setup

4.1 Corpus and split

The corrected procedural corpus contains 24,000 fragments with explicit saved membership: 20,000 training, 2,000 validation, and 2,000 test samples. Generation uses seed 42, 4–16 measures, 2–8 voices, six instrument labels, seven rhythmic profiles, and seven gesture labels. The split summary records `smoke=false`. CPU smoke samples are stored separately and do not enter formal

metrics.

4.2 Training protocol

All formal neural configurations use a six-layer Transformer with hidden size 384, six heads, feed-forward size 1,536, dropout 0.1, and context length 256. Optimization uses AdamW with learning rate 3×10^{-4} , weight decay 0.01, physical batch size 16, gradient clipping at norm 1.0, fp16 autocast, and zero data-loader workers on Windows. Training is limited to 60 epochs with early-stopping patience 10. If CUDA memory is insufficient, the runner retries batch size 8 with two accumulation steps and then batch size 4 with four accumulation steps without reducing the corpus.

Each epoch writes a resumable last checkpoint containing model, optimizer, scaler, random-number states, epoch history, and corpus/configuration hashes. The best validation checkpoint is stored separately. Formal evaluation is permitted only when the checkpoint and completed training summary match the current configuration and processed-corpus hashes.

4.3 Hardware and software gate

The formal target is Windows 11 with an NVIDIA GeForce RTX 4060 Ti 16GB, Python 3.10 or 3.11, and CUDA-enabled PyTorch. The current environment audit reports Python 3.11.9, PyTorch 2.5.1+cu121, CUDA availability, and an NVIDIA GeForce RTX 4060 Ti. These environment facts do not imply that the corrected full training has completed; completion is determined from checkpoints, metric rows, tables, and export artifacts.

4.4 Compared configurations

The full runner evaluates thirteen named configurations: an independent procedural rule reference, a shared neural generator with single-candidate decoding, the same generator with four-candidate constraint-guided decoding, two no-constraint controls, four single-condition removals, and four focused-condition models. All neural condition ablations use the same saved corpus and split. The rule reference generates a fresh deterministic realization for each test condition and never reads the stored target event list.

The proposed and vanilla rows share one trained conditional generator so that their difference isolates candidate count and reranking. Ablation models are trained separately with the specified condition-prefix fields hidden. Metrics for ablations are computed against the unchanged original targets.

4.5 Evaluation and export

Teacher-forced token metrics enumerate all score-body windows in the 2,000-item test split. Generation metrics use 2,000 test conditions per configuration. Twenty examples per evaluated configuration are attempted for MusicXML export, and the final expert package accepts only files that parse structurally and match requested measure and voice counts.

At the time of this revision, the corrected full neural result set is `PENDING_REAL_EXPERIMENT`. Legacy v2 checkpoints and smoke outputs are excluded from the corrected tables and manuscript claims.

5 Evaluation Metrics

Body-token diagnostics. Token accuracy is the fraction of unmasked score-body targets assigned the maximum logit. Cross-entropy is accumulated over the same target tokens across all explicit windows. Condition-prefix and overlap-context positions are masked.

Pitch-class-set measures. For generated support $P(x)$ and nonempty target set S , coverage is $|P(x) \cap S|/|S|$, precision is $|P(x) \cap S|/|P(x)|$, and Jaccard similarity is $|P(x) \cap S|/|P(x) \cup S|$. Interval-vector distance is the L_1 distance between generated and target six-entry interval-class vectors. These metrics are not applicable when no pc-set target exists.

Serial measures. Cyclic row-order accuracy compares generated pitch classes with repeated positions of the requested transformed row. Strict serial-transformation accuracy counts complete non-overlapping twelve-note blocks that exactly equal the requested form. Aggregate completion is the proportion of the twelve pitch classes present and is averaged only over row-conditioned samples.

Rhythmic measures. The density curve counts note onsets per measure. Rhythmic-profile distance compares the generated curve, normalized by requested voice count, with a deterministic profile template available to decoding. Density-curve error compares the generated curve with the stochastic target curve saved during corpus generation. The latter is held out from model conditions and reranking.

Gesture measures. Gesture features include onset dispersion, register spread, note density normalized by requested voices, mean duration, and per-voice rest coverage. Rest coverage is computed from interval unions and is bounded in $[0, 1]$. Feature closeness is averaged into a heuristic gesture consistency score. This score is explainable but is not a perceptual rating.

Range, span, voice, and notation checks. Range-violation rate is the proportion of pitched events outside the requested instrument range. Content-span adherence is the realized event end time divided by the requested duration, capped at one. Voice adherence is the fraction of requested voice indices containing generated material. MusicXML evaluation separately reports structural parse success, requested measure-count adherence, and requested part-count adherence.

Applicability and aggregation. Every aggregate is computed only over samples for which the corresponding target exists. Missing-target values are written as not applicable rather than zero or one. Formal CSV rows must contain all required metric columns and exactly 2,000 evaluated conditions. No corrected numerical comparison is reported until all required configuration rows pass the provenance and completion gates.

6 Results

6.1 Corrected full-run status

The corrected v3 corpus split is available with 20,000 training, 2,000 validation, and 2,000 test fragments and `smoke=false`. The code-level verification suite and CPU smoke workflow pass, and the independent rule reference has been exercised on the full test split. These checks establish pipeline integrity but do not replace neural training.

The corrected neural experiment matrix is not yet complete. Token accuracy, constraint metrics, ablation comparisons, and proposed-versus-vanilla differences therefore remain `PENDING_REAL_EXPERIMENT`. Legacy v2 numerical tables, figures, and controlled-decoding statistics are not used as evidence for the corrected protocol.

6.2 Formal result tables

`PENDING_REAL_EXPERIMENT`: the corrected main-result table has not yet been generated.

`PENDING_REAL_EXPERIMENT`: the corrected ablation table has not yet been generated.

6.3 MusicXML examples

Twenty corrected full-model examples and paired analysis reports will be admitted to the expert package only if each file parses as MusicXML and matches the requested measure and voice counts. The final count and automatic metrics remain `PENDING_REAL_EXPERIMENT` until the proposed checkpoint evaluation completes.

7 Discussion

The corrected design isolates three questions that were confounded in the legacy workflow. First, serial examples and small-set examples now have mutually compatible pitch targets. Second, the single-candidate and guided decoders share one trained generator, so their comparison concerns inference-time selection rather than initialization differences. Third, condition ablations preserve corpus membership and target events while hiding only selected prefix fields.

The revised metric protocol also separates optimization targets from held-out diagnostics. Candidate reranking can use a deterministic rhythmic-profile template, but it cannot inspect the stochastic density curve realized in the corpus target. Pc-set precision supplements recall-like coverage, strict serial-form accuracy is distinct from cyclic row order, and requested score span and voice count are measured explicitly.

These changes improve interpretability but do not establish artistic value. The corpus remains procedural, gesture consistency remains heuristic, and several automatic metrics also participate in candidate selection. Any favorable full-run change will therefore concern implemented symbolic controllability on this testbed. Composer preference, material coherence, engraving usefulness, and transfer to independent scores require blind expert ratings and legally supplied external MusicXML.

No corrected neural result is interpreted here before the formal run completes. Once the v3 CSV and JSON artifacts exist, this section should discuss endpoint-specific gains and trade-offs, not claim uniform superiority.

8 Limitations

The procedural corpus is legal and reproducible but represents one designed distribution rather than the breadth of contemporary compositional practice. A model can learn the generator’s regularities without transferring to independent human-authored scores. Validation on legally supplied MusicXML and blind ratings by composers or post-tonal analysts remain necessary.

Automatic metrics reduce complex musical judgments to inspectable features. Pc-set coverage is recall-like, interval vectors ignore ordering and register, cyclic row accuracy depends on serialized note order, and gesture consistency uses hand-specified feature targets. Strict serial-form accuracy returns zero when a fragment contains no complete aggregate, which distinguishes prefix agreement from complete transformation realization but can penalize intentionally sparse sketches.

The event vocabulary uses a 4/4 sixteenth-note grid and omits tuplets, microtonality, changing meters, extended techniques, detailed articulation, beaming, dynamics in the evaluated configuration, and engraving layout. One instrument label is repeated across parts, so mixed-ensemble orchestration is outside the current interface. Structural MusicXML validity and requested part/measure counts do not establish publication-quality notation.

Condition-preserving windows expose every saved body token but restrict each prediction to a 256-token context. Candidate reranking uses a fixed candidate budget and penalty vector, and most reported constraint metrics overlap with that selection objective. Sensitivity to weights, sampling temperature, candidate count, and alternative symbolic encodings remains to be tested.

Finally, corrected full neural metrics are pending. Smoke outputs and legacy v2 experiments are retained only for engineering history and must not be interpreted as final evidence.

9 Reproducibility Statement

All bundled data are generated from source code and fixed seeds. No copyrighted post-1945 score is downloaded or required. The corrected processed corpus is stored as `data/processed/project2_v3_main.pt`, and its explicit split is summarized in `results/project2_v3_full_split_summary.json`.

On Windows, `scripts/smoke_project2.ps1` runs the isolated CPU verification workflow. The corrected CUDA pipeline is invoked with `scripts/run_project2_full_local.ps1-Resume`. It verifies Python 3.10/3.11, CUDA availability, tests, memory thresholds, corpus counts, configuration hashes, checkpoint hashes, metric-row counts, MusicXML structure, requested measures, requested voices, tables, and expert-package files.

Training writes a best checkpoint and a resumable last checkpoint containing optimizer, scaler, random-number, and epoch state. If CUDA memory is exhausted, the runner reduces physical batch size and increases accumulation without changing the corpus. Every completed training summary records the best epoch, elapsed time, effective batch, OOM adjustment, data/configuration hashes, and measured peak process RAM and allocated VRAM.

Corrected aggregate outputs use the `project2_v3_` prefix until all gates pass. The runner then promotes them to the canonical `results/project2_metrics.csv`, `results/project2_constraints.csv`, `results/project2_generation_examples.json`, `paper/tables/project2_main_results.tex`, `paper/tables/project2_ablation_results.tex`, and `expert_eval/project2/` paths. This delayed promotion prevents smoke or partial runs from replacing final evidence.

The source repository is maintained at the location stated on the title page. Large processed tensors and checkpoints may be excluded from version control, so an archival release must include code, configs, hashes, result tables, and regeneration instructions.

10 Conclusion

This work defines a reproducible score-level workflow for neural-symbolic post-tonal composition assistance. A legal procedural corpus, condition-preserving Transformer training, explicit

pitch-class-set and serial analysis, grammar-constrained candidate generation, MusicXML export, and per-fragment reports are integrated in one local pipeline.

The corrected protocol is designed to distinguish sequence modeling, inference-time reranking, condition ablation, held-out rhythmic evaluation, and notation adherence. Its scientific claims will remain limited to automatic controllability on the synthetic testbed until the full v3 experiment completes. Human evaluation and independent legal MusicXML validation are required before drawing conclusions about artistic usefulness.

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